

MooreVIEW Cloud Phase 1 — Atlanta Deploy (atl1)

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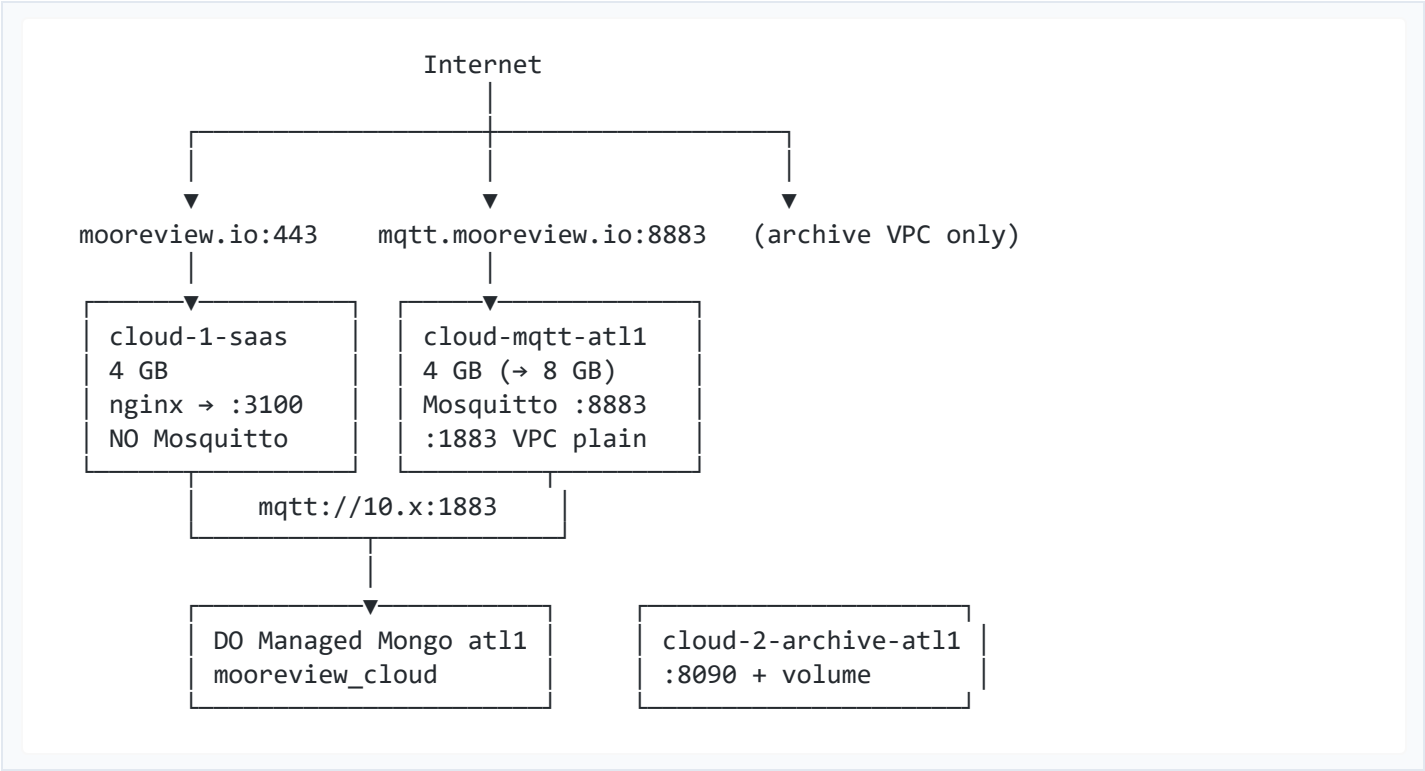
Production (Year 1): DigitalOcean **atl1** — **three-droplet** layout + Managed MongoDB.

Mosquitto lives on **cloud-mqtt-atl1** from day one. Future growth is mostly: add SaaS droplets (+ Load Balancer when >1), bump Managed Mongo tier, optionally resize MQTT RAM — **no broker migration**.

Field broker URL (permanent): `mqtt://mqtt.mooreview.io:8883`

Automated deploy: `npm run deploy:phase1-atl` (after DO resources exist)

1. Architecture



Hostname	Role	Public ports
mooreview.io	SaaS Studio, login, API	443 (80 for certbot)
mqtt.mooreview.io	Field MQTT TLS	8883
Archive	zstd blobs	none (VPC 8090 from SaaS only)

SaaS firewall: 22, 80, 443 — **not 8883**.

MQTT firewall: 22, **8883**.

Never put a Load Balancer on MQTT ingest `:8883` .

2. DigitalOcean resources

Resource	Name	Region	Spec
VPC	mooreview-prod-atl1	atl1	
MongoDB	mooreview-prod-mongo	atl1	10–20 GB → scale tier
Droplet	cloud-1-saas-atl1	atl1	4 GB / 2 vCPU
Droplet	cloud-mqtt-atl1	atl1	4 GB / 2 vCPU
Droplet	cloud-2-archive-atl1	atl1	1–2 GB
Volume	archive-prod-atl1	atl1	100 GB+ → /data/archive
Firewall	fw-saas-atl1	atl1	cloud-1: 22, 80, 443
Firewall	fw-mqtt-atl1	atl1	cloud-mqtt: 22, 8883
Firewall	fw-archive-atl1	atl1	cloud-2: 8090 from SaaS private IP only

3. DNS

Record	Target	When
mqtt.mooreview.io	cloud-mqtt public IP	Before field device onboarding
mooreview.io	cloud-1-saas public IP	After SaaS verification gate
www.mooreview.io	same	Production cutover
test.mooreview.io	NYC sandbox	Independent

4. Secrets (production only)

Generate once (different from sandbox):

```
openssl rand -hex 32 # JWT_SECRET
openssl rand -hex 24 # PLATFORM_ADMIN_KEY
openssl rand -hex 32 # ARCHIVE_SERVER_TOKEN
openssl rand -hex 16 # MOSQUITTO_PASS (shared: mqtt + SaaS hub + field)
```

Secret	Used on
JWT_SECRET	SaaS
PLATFORM_ADMIN_KEY	SaaS admin API
ARCHIVE_SERVER_TOKEN	Archive + SaaS compact job

MOSQUITTO_PASS	MQTT droplet, field devices, SaaS hub
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5. Env files (on droplets)

Droplet	File	Purpose
cloud-mqtt	/etc/mooreview/mqtt.env	MOOREVIEW_DOMAIN=mqtt.mooreview.io, Mosquitto auth
cloud-1-saas	/etc/mooreview/saas.env	Mongo URI, JWT, PUBLIC_* URLs
cloud-1-saas	/etc/mooreview/archive-compact.env	Archive private IP + token
cloud-2-archive	/etc/mooreview/archive.env	ARCHIVE_ROOT, token, port 8090

SaaS MQTT: Do **not** run `enable-saas-mqtt.sh`. Use `configure-saas-mqtt-remote.sh` to set `MOOREVIEW_MQTT_BROKER=mqtt://10.x.x.x:1883`.

Templates in repo: `deploy/cloud/phase1/droplet-*/`

6. Deploy order

6a. Automated (recommended)

After DO console setup (Section 7):

1. Copy `deploy/cloud/phase1/phase1-atl.env.example` → `phase1-atl.local.env`
2. Fill SSH hosts (`ARCHIVE_SSH` , `MQTT_SSH` , `SAAS_SSH`), `MONGODB_URI` , secrets, `PUBLIC_*`
3. Ensure SSH from Windows PC to all three droplets
4. Run: `npm run deploy:phase1-atl`

Automation runs: **archive** → **mqtt** → **saas** → **finish** → **verify**

npm command	Purpose
<code>npm run build:phase1-bundles</code>	Build tgz only
<code>npm run deploy:phase1-atl</code>	Full deploy
<code>npm run deploy:phase1-atl:dry-run</code>	Preview without SSH
<code>npm run deploy:phase1-atl:upload-only</code>	Upload bundles + scripts only

IP-only smoke test (before DNS): set in `phase1-atl.local.env` :

```
RUN_MQTT_CERTBOT=false
RUN_SAAS_CERTBOT=false
```

6b. Manual order (WinSCP fallback)

1. `npm run build:phase1-bundles`
 2. Create VPC, Mongo, firewalls, three droplets, archive volume — all **atl1**
 3. DNS: `mqtt.mooreview.io` → MQTT droplet IP
 4. **cloud-2-archive** — install → record **private IP**
 5. **cloud-mqtt-atl1** — certbot → `install-mqtt-droplet.sh` → record **private IP**
 6. **cloud-1-saas** — `saas.env` → `install-saas.sh` → Mongo allowlist → seed
 7. `configure-saas-mqtt-remote.sh` with MQTT private IP
 8. Verify on IPs — **before** `mooreview.io` DNS
 9. `certbot --nginx` for `mooreview.io` (after DNS)
 10. `enable-phase1-archive-compact.sh`
 11. DNS cutover → update `PUBLIC_*` → restart SaaS
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7. Pre-deploy checklist (DO console)

Automated path

- ☐ DO resources created (VPC, Mongo, droplets, firewalls, volume)
- ☐ `phase1-atl.local.env` filled (SSH, Mongo URI, secrets)
- ☐ DNS `A mqtt.mooreview.io` → MQTT public IP (or `RUN_MQTT_CERTBOT=false`)
- ☐ SSH works to archive, mqtt, saas from deploy PC
- ☐ `npm run deploy:phase1-atl`
- ☐ Mongo trusted source = SaaS **public** IP

VPC and Mongo

- ☐ VPC: `mooreview-prod-atl1` (region atl1)
- ☐ Managed Mongo: `mooreview-prod-mongo` , DB `mooreview_cloud`
- ☐ Mongo automated backups enabled
- ☐ Connection string saved

cloud-2-archive-atl1

- ☐ Droplet 1–2 GB, same VPC
- ☐ Volume 100+ GB mounted at `/data/archive`
- ☐ Firewall: 8090 from SaaS private IP only
- ☐ `curl http://127.0.0.1:8090/health` OK
- ☐ Private IP recorded: 10._____

cloud-mqtt-atl1

- ☐ Droplet 4 GB, same VPC
- ☐ Firewall: 22, 8883

- ☐ certbot for `mqtt.mooreview.io` (if not automated)
- ☐ `mosquitto_pub` local test OK
- ☐ Private IP recorded: 10._____

cloud-1-saas-atl1

- ☐ Droplet 4 GB, same VPC
- ☐ Firewall: 22, 80, 443 only (**no 8883**)
- ☐ `install-saas.sh` — **no** `enable-saas-mqtt.sh`
- ☐ Mongo allowlist includes SaaS public IP
- ☐ `configure-saas-mqtt-remote.sh` with MQTT private IP
- ☐ `curl http://127.0.0.1:3100/health` (hub connected)

Archive compact (SaaS)

- ☐ `/etc/mooreview/archive-compact.env` configured
- ☐ `enable-phase1-archive-compact.sh`
- ☐ Compact dry-run OK

8. Verification gate

MQTT droplet:

```
mosquitto_pub -h 127.0.0.1 -u mooreview -P '...' -t test -m ok
systemctl status mosquitto
```

SaaS droplet:

```
curl -s http://127.0.0.1:3100/health
grep MOOREVIEW_MQTT_BROKER /etc/mooreview/saas.env
systemctl status mooreview-saas
! systemctl is-active mosquitto && echo "OK: no local mosquitto"
```

End-to-end (SaaS via VPC):

```
mosquitto_pub -h 10.x.x.x -u mooreview -P '...' \
  -t 'mooreview/v1/demo-tenant/test_device/telemetry' \
  -m '{"deviceId":"test_device","tags":[{"id":"T1","type":"BOOL","value":true}]}'
```

Post-deploy:

- ☐ `mqtt.mooreview.io:8883` TLS reachable

- ☐ SaaS login on droplet IP — demo tenant
- ☐ No mosquitto on SaaS droplet
- ☐ Archive reachable from SaaS (private IP)

9. DNS cutover (production)

- ☐ A `mooreview.io` → cloud-1-saas public IP
- ☐ `PUBLIC_APP_URL` / `PUBLIC_API_URL` → `https://mooreview.io`
- ☐ `systemctl restart mooreview-saas`
- ☐ `https://mooreview.io/login` verified
- ☐ Field provisioning: `mqtt://mqtt.mooreview.io:8883`

10. Troubleshooting

Symptom	Fix
SSH hangs on deploy	Set <code>SSH_KEY</code> in config; test <code>ssh -o ConnectTimeout=15 root@IP</code>
certbot fails on mqtt	Point DNS first, or <code>RUN_MQTT_CERTBOT=false</code>
SaaS health OK, no MQTT hub	Re-run deploy <code>-Step finish</code> ; check <code>MOOREVIEW_MQTT_BROKER</code>
Compact dry-run fails	Add SaaS public IP to Mongo allowlist
Local mosquitto on SaaS	Re-run <code>configure-saas-mqtt-remote.sh</code>

11. Future scaling (summary)

Stays fixed	Grows with fleet
<code>mqtt.mooreview.io</code> hostname	+ SaaS droplets + LB on :443
Archive volume + compact pipeline	Managed Mongo tier/disk
VPC <code>mooreview-prod-atl1</code>	Ingest shards (:3090) → same MQTT VPC broker

Do not redo: field broker URL, Mongo URI pattern, archive pipeline, MQTT topic layout.

12. Cost reference (atl1)

Layout	~Monthly
2-droplet (MQTT on SaaS)	~\$70–90
3-droplet (+ dedicated MQTT 4 GB)	~\$95–115

See also: docs/CLOUD_VERTICAL_SHARDS.md · docs/marketing/CLOUD_INFRA_PRICING.md

Repo references

Path	Purpose
docs/CLOUD_DEPLOY_DO_PHASE1_ATL-MQTT.md	Architecture (markdown)
deploy/cloud/phase1/DEPLOY-ATL-MQTT-AUTOMATED.md	Automated runbook
deploy/cloud/phase1/CHECKLIST-ATL-MQTT.txt	Printable checklist
deploy/cloud/phase1/phase1-atl.env.example	Deploy config template
scripts/deploy-phase1-atl-mqtt.ps1	Windows orchestrator

Sandbox: nyc1 @ test.mooreview.io — separate from production atl1.